

SOLVABLE GROUPS

See also [METABELIAN GROUPS](#) and [NILPOTENT GROUPS](#).

(S1) (A.I.Mal'cev) Describe the automorphism group of a free solvable group of finite rank. In particular, is this group finitely generated? [Background](#)

(S2) (M.I.Kargapolov) The word problem for groups admitting a single defining relation in the variety of all solvable groups of a given derived length. [Background](#)

(S3) (M.I.Kargapolov) Is it true that every group of rank > 2 admitting a single defining relation in the variety of all solvable groups of a given derived length, has trivial centre? [Background](#)

(S4) (M.I.Kargapolov) Is there a number $N = N(k,d)$ so that every element of the commutator subgroup of a free solvable group of rank k and derived length d , is a product of N commutators? [Background](#)

(S5) (P.M.Neumann) Is it true that if A, B are finitely generated solvable Hopfian groups, then $A \times B$ is Hopfian?

(S6) (V.N.Remeslennikov) The conjugacy problem for finitely generated abelian-by-polycyclic groups.

(S7) (D.Robinson) Is there a finitely presented solvable group satisfying the maximum condition on normal subgroups, with unsolvable word problem ?

(S8) (G.Baumslag, V.Remeslennikov) Is a finitely generated free solvable group of derived length 3 embeddable in a finitely presented solvable group?

*(S9) (B.Fine, V.Shpilrain) Let u be an element of a group G . We call u a *test element* if, whenever $f(u)=u$ for some endomorphism f of the group G , this f is actually an automorphism of G . Does the free solvable group of rank 2 and derived length $d>2$ have any test elements? [Background](#)